Biospectra : ISSN: 0973-7057, Vol. 17(2), September, 2022, pp. 79-84
An International Biannual Refereed Journal of Life Sciences

14(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 616 www.mjl.clarivate.com

Utility of *Azolla* as animal feed

Naina Srivastava*
Department of Botany, D.A.V. (P.G.) College, Karanpur, Dehradun, Uttarakhand, India

Received : 17th May, 2022 ; Revised : 18th June, 2022

Abstract- Various species of *Azolla* plants are highly productive, with high in protein and minerals, and other nutrients. The special thing is that *Azolla* is a portion of tasty and nutritious food for cows, buffaloes, chickens, pigs, goats, ducks, and animals. Along with this, apart from adequate amounts of vitamin A and vitamin B12, essential minerals like iron, calcium, magnesium, phosphorus, copper, manganese, etc. are also found in it. It contains almost all the essential amino acids, probiotics, bio-polymers, and beta carotene. If you look at the cost of *Azolla*, there is more profit in very less expenditure, because there is no need to invest much money in it. Farmers can grow it in a shady place in their home and farm, as it requires only one plastic sheet, for laying in the bed. *Azolla* is a treasure trove of complete nutrients, from which all the nutrients of animals are fulfilled. Regular feeding of *Azolla* to the animals increases their production as well as increases their body and keeps the animals healthy. The maximum benefit has been found in the cow because the cow does not eat leather, clothes, soil, etc. The main reason for this is that if the cow is not fed balanced fodder, there is a lack of nutrients in her body, due to which she eats leather, slippers, clothes, and polythene. It was seen in the survey that the cow which was fed *Azolla* daily does not eat all of these.

Key words: *Azolla*, vitamin, iron, calcium, magnesium, phosphorus, copper, manganese

INTRODUCTION

Azolla (*Azolla* sp.) is a free-floating aquatic fern plant with beautiful fine leaves. *Azolla* is found in ponds, water pits, and moist-warm tropical regions.[1] It requires about 25-50 percent light and very sensitive to the availability of water and cannot thrive without water. For its reproduction and development, an optimum temperature of 20-30 degrees centigrade is required and relative humidity of 95-90 percent is considered good. Its optimum pH value is 5-7. *Azolla* obtains essential nutrients from the water itself.[2] Due to its nutritional properties, it is considered to be an excellent alternative to the skin of animals. It gives better results at less cost and is ready in a very short time. The most important thing, there is no need for separate ground in preparing it from it.[3] That is not the case, you can put your eyes on the courtyard. Therefore, *Azolla* is full of versatility, high quality, and nutritious elements, nothing less than a boon for the farmers.[4]

*Corresponding author :
Phone : 9105897878
E-mail : drnainasrivastava@gmail.com

MATERIALS & METHODS

Selection of pond: It is better to choose a pond near the house for *Azolla* for the necessary care. A regular water supply is essential. Partial sunlight at the place where the pond is selected will ensure better yield, otherwise, the shade will have to be provided.[5] This will not only reduce the evaporation of water but will also increase the yield of *Azolla*.[6]

Pond size and construction: The size of the pond will depend on the number of animals. 0.7 kg per day for small cattle holders.[7] An area of 6 by 4 feet is sufficient for the cultivation of *Azolla* to produce more supplementary fodder. The selected area should be perfectly flat.[1] The walls of the pond can be built either with bricks or with excavated soil. After laying the durable plastic sheet in the pond, all the walls should be secured by placing bricks on the walls.[8] To prevent leakage of water, there should be no holes or cracks and the pond should be covered with a net mesh cover.[9] This can provide partial shade and keep leaves and other debris in the pond.

Pond made of readymade High-Density Polyethylene (HDPE) or Poly Vinyl Chloride (PVC): These ponds come in different sizes like 10 by 3, 15 by 3, 20 by 3 feet, etc. Farmers can choose the size of the pond based on the number of animals available with them and the daily requirement of *Azolla,*[10] which they need daily for feeding. About 2 kg per day from a pond of 20 by 3 feet. Fresh *Azolla* can be produced. These HDPE ponds last longer as compared to ponds made of polythene sheets.[11]

***Azolla* Production:** Fertile mixed soil with dung and water needs to be mixed evenly in the pond. This is necessary for a pond of size 6 by 4 feet. It should be given evenly throughout the pond. The depth of water in the pond should be 4-6 inches. The surface of the pond should also be level so that the water depth remains the same throughout the pond area.[12] During the monsoon season, rainwater can be harvested from rooftops, etc. If this stored water is used for the cultivation of *Azolla*, it will ensure excellent and rapid growth.[13]

Pond maintenance: One kg. Use of cow dung and 100 grams of superphosphate once in 15 days in the pond will lead to better growth of *Azolla*. The pond should be emptied once in six months. More *Azolla* can also be used in the fields.[14]

***Azolla* evacuation:** *Azolla* production is completed in three weeks. It can be removed daily after full development. In case of high production of *Azolla*, it can be dried in shade. It can be safely preserved for future use.[15] In a study that fed animals of more than 100 dairy farmers from different villages of Chitradurga district of Karnataka under the NAIP Aajeevika Project, feeding *Azolla* on an average (fresh weight) of 700 grams per cow per day improved the monthly milk production to 10 liters per day. The product is obtained from the cow. It takes a few days for animals to get used to its taste.[16] The grains should be thoroughly washed with fresh water in the initial stages for better utilization. When there is a high production, *Azolla* can be dried in the shade and stored in plastic drums. Dried *Azolla* can be used in times of shortage of animal feed.

AZOLLA AS ANIMAL FEED

Azolla was chosen especially because it has been used as a balanced diet for animals. It is also rich in protein, minerals, and amino acids.[17] Feeding *Azolla* not only keeps the animal healthy and healthy but has also been found to increase the productivity of milk by 10-15 percent. Apart from this, it has also proved to be very helpful in physical development.[18] A unit for each of *Azolla's* demonstrations has been developed at all the three selected centers.

***Azolla* is rich in nutrients:**

Essential nutrients are also available in *Azolla*. It contains about 25 to 30 percent protein, which is a major source of lysine, arginine, and methionine.[19] Due to the small amount of lignin in *Azolla*, digestion becomes easier in the body of animals. The use of *Azolla* in place of conventional fertilizer urea increases production. Nitrogen 28 to 30 percent, minerals 10 to 15 percent, beta carotene calcium, magnesium, potassium, phosphorus, iron, and copper are found in abundance in this.[20] *Azolla* is also known as Green Gold based on the availability of nutrients. *Azolla* Utility For good milk production, it is necessary that milk production should be increased at less cost. From this point of view, *Azolla* is easy, cheap, and beneficial in the form of fodder. We have only 5.25 percent of land available for fodder production. The production of *Azolla* is very simple, which is used in the form of food for cows, buffalo, goats, pigs, poultry, etc. It can be easily grown in ponds, rivers, pits, tubs, etc.

Table 1- Chemical composition of *Azolla* (on a dry weight basis)

Sl.No.	Nutrients	Quantity in percentage
1.	protein	22.5
2.	fiber	12.5
3.	fat	03.2
4.	Carbohydrate	50.0
5.	calcium	1.16
6.	Phosphorus	1.29
7.	magnesium	0.35
8.	manganese	174.42
9.	zinc/zinc	87.59
10.	copper/copper	16.74

Souse: http//www.vikashpedia.in

It is known from Table-1 that *Azolla* contains mineral elements like calcium, phosphorous, manganese, zinc and copper, etc. Apart from this, Vitamin A and Vitamin B, are found in plenty in it. As well as all of it. Essential amino acids are insufficient in quantity.[21] The deficiency of protein and green fodder can be met by using *Azolla* in the form of cheap organic supplement ration in the diet of milch animals like cows, buffalo, sheep, goats, etc.

Profit-Cost Analysis of *Azolla*:

If you look at the cost of *Azolla*, there is more profit in very less expenditure, because there is no need to invest much money in it.[22] Farmers can grow it ina shady place in their home and farm, as it requires only one plastic sheet, for laying in the bed. If seen from *Azolla*, a net profit of about Rs 37,500 can be obtained per annum.[23]

Table 2- Increase in production and benefits of feeding *Azolla* to animals

Animal	Amount	increase body weight percentage / output
Cow	1.5 kg daily	15 percent
Buffalo	1.5 kg daily	12 percent
She goat	1.0 kg daily	15.9 percent
lamb	1.0 kg daily	16.0 percent
chicken	150 grams per day	12.5 percent
Pig	2.0 kg daily	20.0 percent
Horse	1.5 kg daily	11.5 percent

Souse: http//www.vikashpedia.in

Along with this, the production method of *Azolla* is so easy and simple that even the women of the house can produce it. By using it for cattle and small animals like goats, chickens, ducks, and pigs, you can get a good income by increasing their average weight in a short time. The use of *Azolla* in paddy fields as organic manure has been found, as a result of which organic carbon in the soil has increased and soil fertility has increased.[24] It has proved helpful in making agriculture a long-term and pure environment.

Benefits of *Azolla* as Animal feed

Azolla is a treasure trove of complete nutrients, from which all the nutrients of animals are fulfilled. Regular feeding of *Azolla* to the animals increases their production as well as increases their body and keeps the animals healthy.[25] The maximum benefit has been found in the cow because the cow does not eat leather, clothes, soil, etc. The main reason for this is that if the cow is not fed balanced fodder, there is a lack of nutrients in her body, due to which she eats leather, slippers, clothes, and polythene.[26] It was seen in the survey that the cow which was fed *Azolla* daily does not eat all of these. The effect of this on animals and the body of animals are given in table-2.

To increase the milk production of animals, feed the animals by applying *Azolla* fodder, this will not only increase the milk production but also reduce the cost. At present, *Azolla* can be called a good nutritional option for milch animals, chickens, and goats, considering the availability of useful nutrients for animals.[27] It has also become a good source of green fodder due to its unique quality of giving more production in less time. Due to climate and climate not having a special effect on *Azolla* production, it can be produced in all parts of the country.[16] Farmers can double their income by producing *Azolla* on their own with general guidance. It would also be relevant to mention here that there is also a reduction in the infertility of *Azolla* animals.[3]

It is being used to increase the quantity and fat percentage of milk in animals because the cost of its production is less. Therefore, its utility is increasing day by day. *Azolla* nutrition increases milk production by 10 to 20 percent.[28] By using up to 60 grams, the concentration of food can be reduced by 10 percent. In crossbreed cow, 2 kg. Instead of concentrated food, 2 kg. If *Azolla* is fed, then the expenditure can be reduced by about 35 to 40 percent by combining both milk production and labor.[29] *Azolla* can be given directly to livestock in the ratio of 1:1 along with ration. In this way, if we see, *Azolla* increases the milk of milch animals in the right amount, and income increases with less expenditure. If the farmer's crossbreed cow is going to give 10 liters of milk, then he feeds it with a concentrated feed of 5.5 kg. will give If the price of food is Rs. 20 per kg. If kept, Rs. 110 per kg per day.[30] If kept, then about 3 to 3.50 kg in it. *Azolla* can be used instead of concentrate, which costs 65 paise per kg. If so, then using *Azolla* will cost 3 rupees for that animal. In this way, instead of Rs 70, there will be an increase in milk production by about 2 liters in just about Rs 3. Due to this, his total profit will be 2 liters of milk and 70 rupees ration and wages from all three.[22] In this way, farmers can more than double their income.

CONCLUSION

Azolla is a member of the Azollaceae family, which is a water-growing fern. It is only living member of this

family. It can be grown easily in moist, humid, and hot climates. *Azolla* uses atmospheric nitrogen for its growth, which is stored in the plant. It is being produced in many areas in South India, especially in Tamil Nadu and Kerala and Krishi Vigyan Kendra, Pali, Ajmer, and Jaipur. *Azolla* is a low-cost, nutritious food for livestock. Based on dry weight, 25-35 percent protein, 10-12 percent minerals, and 7-10 percent amino acids are found in it. It is a fast-growing variety. Its production starts within 8-10 days after sowing. Cow, buffalo, chicken, sheep, goat eat *Azolla* with great fervor, the air is easily digested, as a result of which a 10-15 percent increase in milk production has been found, and by feeding regularly for one month, goat, pig, chicken About 25-30 percent increase in weight has been found. An increase of 25-30 percent was found in the production and productivity of paddy by using *Azolla* with blue-green algae (cyanobacteria) in the paddy field.

REFERENCE

1. **Shukla M., Bhattacharyya A., Shukla P. K., Roy D., Yadav B., & Sirohi R. 2018.** Effect of *Azolla* feeding on the growth, feed conversion ratio, blood biochemical attributes and immune competence traits of growing turkeys. *Veterinary world*. **11(4):** 459–463.

2. **Agnihotri N. 2018.** Effect of Pollution on Human and Environment. *Trends in Biosciences.* **11(25):**3411-3417

3. **Samad F., Idris L. H., Abu Hassim H., Goh Y. M., & Loh T. C. 2020.** Effects of *Azolla* spp. as feed ingredient on the growth performance and nutrient digestibility of broiler chicken. *Journal of Animal Physiology and Animal Nutrition*. **104(6):** 1704–1711.

4. **Agnihotri N. 2019.** Hazards of Chromium Pollution and Its Prevention. *Bulletin of Environment Pharmacy & Life Sciences.* **8(8):**1-5

5. **Brouwer P., Schluepmann H., Nierop K. G., Elderson J., Bijl P. K., van der Meer I., de Visser W., Reichart G. J., Smeekens S., & van der Werf A. 2018.** Growing *Azolla* to produce sustainable protein feed: the effect of differing species and CO_2 concentrations on biomass productivity and chemical composition. *Journal of the Science of Food and Agriculture*. **98(12):** 4759–4768.

6. **Mishra D. B., Roy D., Kumar V., Bhattacharyya A., Kumar M., Kushwaha R., & Vaswani S. 2016.** Effect of feeding different levels of *Azolla pinnata* on blood biochemicals, hematology and immuno-competence traits of Chabro chicken. *Veterinary World*. **9(2):** 192-198.

7. **Agnihotri N., Sharma T. K. and Sippy D. 2020.** Population Growth Food Security Biodiversity and Covid-19. *The Biobrio*. **8(3-4):**503-508

8. **Agnihotri N. 2020.** Corona Virus: A New Health Hazard in Processing on National conference on Recent Trend and New Frontiers Biotechnology, Agriculture Science and Environmental. *Janome Biotech Publication, Mathura*. **2:**102-103

9. **Shambhvi Katoch S., Chauhan P., & Mane B. G. 2020.** Effect of feeding *Azolla pinnata* in combination with direct-fed microbial on broiler performance. *Tropical animal health and production*. **53(1):** 5.

10. **Chichilichi B., Mohanty G. P., Mishra S. K., Pradhan C. R., Behura N. C., Das A., & Behera K. 2015.** Effect of partial supplementation of sun-dried *Azolla* as a protein source on the immunity and antioxidant status of commercial broilers. *Veterinary World*. **8(9):** 1126–1130.

11. **Acharya P., Mohanty G. P., Pradhan C. R., Mishra S. K., Beura N. C. & Moharana B. 2015.** Exploring the effects of inclusion of dietary fresh *Azolla* on the performance of White Pekin broiler ducks. *Veterinary World.* **8(11):** 1293–1299.

12. **Agnihotri N and Singh A. K. 2021.** Impact of Lockdown on Environment. *Advance Plant Science*. **1(12):**24-29

13. **Abdelatty A. M., Mandouh M. I., Mohamed S. A., Busato S., Badr O., Bionaz M., Elolimy A. A., Moustafa M., Farid O. & Al-Mokaddem A. K. 2021.** *Azolla* leaf meal at 5% of the diet improves growth performance, intestinal morphology and p70S6K1 activation, and affects cecal microbiota in broiler chicken. *Animal : An International Journal of Animal Bioscience*. **15(10):** 100362.

14. **Agnihotri N. 2022.** Utility of Self-Grown Medicinal Plants As Traditional Remedy In Farrukhabad District. *European Journal of Agricultural and Rural Education (EJARE),* Vol. 3 No. 2; 79-86. Available Online at: https://www.scholarzest.com

15. **Kumar I and Gautam M. 2022.** Enhance the Nutritive Value of Diet through Dietary Diversity in the Rural area of Uttar Pradesh: an intervention-based study. *Indian Res. J. Ext. Edu.* **22(2):**29-33

16. **Islam M. A. & Nishibori M. 2017.** Use of multivitamin, acidifier and *Azolla* in the diet of broiler chickens. *Asian-Australasian Journal of Animal Sciences*. **30(5):** 683–689.

17. **Tran T., Miranda A. F., Abeynayake S. W., & Mouradov A. 2020.** Differential Production of Phenolics, Lipids, Carbohydrates and Proteins in Stressed and Unstressed Aquatic Plants, *Azolla filiculoides* and *Azolla pinnata*. *Biology*. **9(10):** 342.

18. **Kumar I. & Gautam M. 2022.** Determinants of Dietary Diversity Score for the Rural Households of Uttar Pradesh State. *Int. J. Food Nutr Diet*. **10(1):**9–16.

19. **Pereira A. L., Monteiro B., Azevedo J., Campos A., Osório H., & Vasconcelos V. 2015.** Effects of the naturally-occurring contaminant microcystins on the *Azolla* filiculoides-Anabaena azollae symbiosis. *Ecotoxicology and Environmental Safety*. **118:** 11–20.

20. **Qiu Y. L., & Yu J. 2013.** *Azolla*-a model organism for plant genomic studies. *Genomics, Proteomics & Bioinformatics*. **1(1):** 15–25.

21. **Rahmah S., Nasrah U., Lim L. S., Ishak S. D., Rozaini M., & Liew H. J. 2022.** Aquaculture wastewater-raised *Azolla* as partial alternative dietary protein for *Pangasius* catfish. *Environmental research*, **208:**112718.

22. **Miranda A. F., Biswas B., Ramkumar N., Singh R., Kumar J., James A., Roddick F., Lal B., Subudhi S., Bhaskar T. & Mouradov A. 2016.** Aquatic plant *Azolla* as the universal feedstock for biofuel production. *Biotechnology for Biofuels*. **9:** 221.

23. **Kumar I. & Gautam M. 2021.** Correlation between Individual Dietary Diversity Score and Nutrients Adequacy Ratio in the Rural Community. *SPR*. **1(4):**258–263.

24. **Banach A. M., KuŸniar A., Grzdziel J., &Woliñska A. 2020.** *Azolla filiculoides* L. as a source of metal-tolerant microorganisms. *PloS one*. **15(5):** e0232699.

25. **Ram N., Ram L., Verma S. K., Agnihotri N., Choudhary M., Chamoli M., Sharma J., Maheshwari R. K. 2022.** To study physicochemical parameters of potable water sources of Agra city (Uttar Pradesh) India. *International Journal of Multidisciplinary Research & Development.* **9(3):**1-7

26. **Azab E., & Soror A. S. 2020.** Physiological Behavior of the Aquatic Plant *Azolla sp.* in Response to Organic and Inorganic Fertilizers. *Plants* (*Basel, Switzerland*). **9(7):** 924.

27. **Bhuvaneshwari K. & Singh P. K. 2015.** Response of nitrogen-fixing water fern *Azolla* biofertilization to rice crop. *3 Biotech*. **5(4):**523–529.

28. **Erickson P. S. & Kalscheur K. F. 2020.** Nutrition and feeding of dairy cattle. *Animal Agriculture*. 157–180.

29. **Brouwer P., Bräutigam A., Külahoglu C., Tazelaar A., Kurz S., Nierop K., van der Werf A., Weber A., & Schluepmann H. 2014.** *Azolla* domestication towards a biobased economy?. *The New Phytologist*. **202(3):**1069–1082

30. **Temmink R., Harpenslager S. F., Smolders A., van Dijk G., Peters R., Lamers L., & van Kempen M. 2018.** *Azolla* along a phosphorus gradient: biphasic growth response linked to diazotroph traits and phosphorus-induced iron chlorosis. *Scientific Reports*. **8(1):** 44-51.

***